Lakshminath Reddy Alamuru

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Summary: Embedded Software Engineer with 4+ years of experience at **Qualcomm, Nokia, and Harman**, specializing in Linux systems, telephony frameworks, and embedded platforms. Skilled in C/C++, Python, AOSP, HAL, and multithreaded programming with a proven record of optimizing performance, developing and integrating APIs, and enhancing reliability across Android and other embedded devices.

SKILLS

Programming Languages: C++, C, Python, Android Java, Verilog, Assembly, Bash.
Frameworks & Modules: Networking Operating System(SRLinux), AOSP, QCRIL, QtiDailer, HALs, Vendor IMS, Yocto, oFono, Exynos-RIL.
Software Technologies & Tools: Embedded Linux, Android SDK, ADB, CMake, Yocto, Bitbake, Git, Modem-Simulator, QXDM, LAF, YANG, gRPC, Docker, BSP, Device Drivers, GDB, Wireshark, ModelSim, Visual Studio, Jira, Confluence.
Others: OOPS Concepts, Design Patterns, Multithreaded, 3GPP specification, SIP protocol, TCP/IP, SOC, SPI, I2C, Embedded Systems, Agile, Debugging, Code Review, Root Cause Analysis.

WORK EXPERIENCE

Nokia, *Sunnyvale, California, USA*
Embedded Software Engineer Intern 01/2025 - 08/2025

- Engineered a **signal-safe alternative** to **libc's localtime_r()** with a timezone caching layer, eliminating tzset()-related deadlocks and improving signal handling reliability by 100% in production.
- Built and integrated **core-level CPU monitoring** in the Linux Platform module, exposing **per-core utilization** across 64+ cores via gRPC/SRCLI using the /proc interface, with Protobuf-based IPC over IDB, enhancing real-time system observability and diagnostics.
- Improved generation of **tech support files in SRLinux** to include the status of the component, reducing the troubleshooting time by 30%.
- Diagnosed and resolved **15+ critical** bugs in **SR Linux's Application** and **Linux Manager** modules by enhancing overall platform reliability.

Qualcomm, *Hyderabad, India*
Embedded Software Engineer 03/2022 - 01/2024

- Led **convergence** of 32+ **PS and CS** telephony APIs in **QCRIL**, improving code maintainability by 45% and ensuring robust end-to-end testing on live network devices
- Optimized and expanded the capabilities of QtiDialer, IMS settings, and IMS features (**Voice, VoLTE, VoWiFi, conference calls**) on **Snapdragon chipsets for Android 13 and 14**, improving call reliability by 25%.
- Collaborated with Google to identify and resolve 15+ dial-up issues, enhancing user experience and compatibility across OEM devices.
- Enhanced vendor IMS, AOSP framework and QCRIL by skillfully resolving potential issues related to **4G/5G Conference calls, DSDA, and upgraded / downgraded video calls** for various operators, leading to a decrease in bug reports 30%.
- Added **simulated functionality for RTT, DSDA**, and call forwarding in **Modem Simulator** using Python, Tkinter, and gRPC, enabling interaction between the simulator UI and HAL (RIL) layer, accelerated feature validation by 40% and reduced manual testing efforts
- Developed and implemented IMS scenarios for the **Log Analysis Framework**, reduced reported issue analysis efforts by 38% for developers.

Harman International(A Samsung Company), *Bangalore, India*
Associate Embedded Software Engineer 10/2020 - 03/2022

- Designed and implemented a streamlined module for **automatic online modem registration**, reducing registration time by 5 seconds with cross-functional collaboration, and ensured seamless integration with the telematics control unit.
- Developed and maintained **4 bootup modules**, reducing modem initialization time by 15% and ensuring 99.9% uptime for RIL daemon operations, enabling seamless modem registration with the oFono framework for stable online connectivity across multiple devices.
- Served as primary liaison for the telephony team during **device bring-up**, resolving 95% of critical issues within deadlines and coordinating with Security, Power, and Kernel teams to ensure the on-time launch of devices
- Stabilized **oFono middleware** and oFono applications concerning **5G/4G data, Voice Calls, SMS, and SIM**, resolving 28% of critical issues.
- Developed and maintained Bitbake files for telephony modules, ensuring the separation of the D-Bus system bus and session bus for enhanced modularity and security.

DEVELOPMENT PROJECTS

**Automonous Plant Simulation — ***C++, Multi-Threading, Parallel Programming* 04/2024 - 05/2024

- Designed and implemented a weighted random part selection algorithm based on time-to-buffer-capacity ratios, optimizing buffer occupancy, minimizing worker thread idle time by 27%.
- Executed a dynamic time out assignment strategy for worker threads based on real-time analysis of buffer area and product orders by decreasing the wait time to less than 5 ms.

**Linux Character Device Driver Development — ***Embedded C, Linux, Kernel Programming* 08/2024 - 10/2024

- Implemented a character device driver in an embedded Linux environment that provides applications running in user space access to multiplication peripheral.
- Developed and tested a Linux app that uses the device driver, ensuring reliable operations and verified functionality.

**Unauthorized Access Point Detection — ***Python, Jupyter Notebook, Numpy, Pandas, Scikit-Learn* 03/2020- 05/2020

- Executed a Machine Learning initiative to detect unauthorized wireless access points, using MLP, KNN, and SVM algorithms. Achieved 84.5% accuracy in identifying rogue connections, enhancing network security, and reducing threats.
- The model was trained using a data set comprising 100,000+ connection requests from 700 devices.

EDUCATION

Master of Science in Computer Engineering 01/2024 - 12/2025
Syracuse University, Syracuse, NY, USA GPA - 3.919/4.0
CourseWork: ***Advanced Data Structures, Operating Systems, Computer Architecture, Object-Oriented Design, System On Chip, Computer Security***

Bachelor of Technology in Electronics and Communication Engineering 08/2016 - 06/2020
National Institute of Technology, Jamshedpur, India